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Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	12	Academic Year :	2026-27

Task - 12

Aim

To establish a Continuous Integration (CI) pipeline using GitHub Actions, automating static code analysis (npm run lint), backend integration testing against an ephemeral MongoDB 7.0 service container (npm test), and production frontend build verification on every push and pull request.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. .github/workflows/ci.yml: Multi-job pipeline triggered on push & pull_request to main.
2. test-backend job: Configures official mongo:7.0 service container with health checks.
3. build-frontend job: Enforces clean production bundle compilation with Vite.

```
// .github/workflows/ci.yml
jobs:
  lint:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: npm --prefix frontend ci && npm --prefix frontend run lint
  test-backend:
    runs-on: ubuntu-latest
    services:
      mongodb:
        image: mongo:7.0
        ports: [ 27017:27017 ]
    steps:
      - run: npm --prefix backend ci && npm --prefix backend test
Output
```

Output

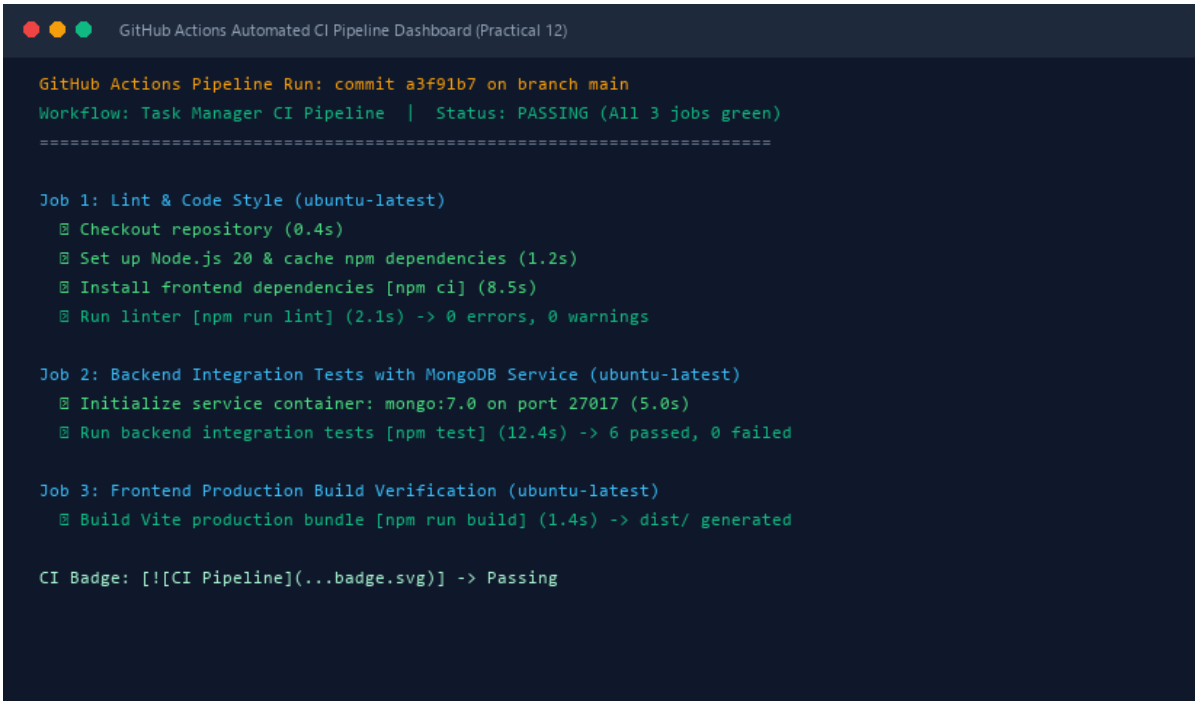


Figure 12: Execution Output & Functional Verification for Practical 12